

Phase I/II Operating Characteristics

TITE versus IPDE-first non-TITE: one-stage and two-stage allocation

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 1						
DLT rate	0.3	0.35	0.4	0.45	0.5	
Efficacy rate	0.05	0.06	0.07	0.08	0.09	
OBD selection (%)						
TITE one-stage	8.5	4.2	3.7	0.9	0.4	82.3
IPDE-first non-TITE one-stage	8	6.1	2.9	1.6	0.2	81.2
TITE two-stage	5	6.8	4.4	1.2	0.2	82.4
IPDE-first non-TITE two-stage	8.5	6	2.9	1.5	0.2	80.9
Mean administrations by dose						
TITE one-stage	8.3	5.1	2.1	1.2	0.4	
IPDE-first non-TITE one-stage	10.6	4.7	2.7	1.2	0.3	
TITE two-stage	8.7	3.3	1.5	0.6	0.2	
IPDE-first non-TITE two-stage	10.5	4.8	2.8	1.2	0.4	
Mean trial duration (days)			Mean duration (days)			
TITE one-stage			636.6			
IPDE-first non-TITE one-stage			846			
TITE two-stage			530.9			
IPDE-first non-TITE two-stage			848.7			

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	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 16						
DLT rate	0.02	0.06	0.09	0.14	0.26	
Efficacy rate	0.2	0.2	0.2	0.2	0.2	
OBD selection (%)						
TITE one-stage	31.2	24.1	16.7	13.9	13.7	0.4
IPDE-first non-TITE one-stage	23	22.8	19.8	17.6	15.5	1.3
TITE two-stage	12.6	18.2	21.5	24.4	23	0.3
IPDE-first non-TITE two-stage	22.4	22	19.5	18.7	16	1.4
Mean administrations by dose						
TITE one-stage	4.2	5	6.2	8.2	6.3	
IPDE-first non-TITE one-stage	7.1	6.6	5.9	5.5	4.9	
TITE two-stage	9	7.1	5.4	4.6	3.7	
IPDE-first non-TITE two-stage	6.9	6.3	6.1	5.7	5	
Mean trial duration (days)			Mean duration (days)			
TITE one-stage			1099.6			
IPDE-first non-TITE one-stage			1246.6			
TITE two-stage			1068.8			
IPDE-first non-TITE two-stage			1248.2			

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	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 20						
DLT rate	0.07	0.14	0.31	0.41	0.52	
Efficacy rate	0.05	0.25	0.25	0.25	0.25	
OBD selection (%)						
TITE one-stage	6.6	40.4	29.5	13.6	2.1	7.8
IPDE-first non-TITE one-stage	4.3	43.7	30.9	12.9	2.7	5.5
TITE two-stage	4.1	36.4	32.6	14.5	3.8	8.6
IPDE-first non-TITE two-stage	5.6	44.4	30.2	12.1	2.7	5
Mean administrations by dose						
TITE one-stage	4.3	9.4	8	5.4	1.8	
IPDE-first non-TITE one-stage	5.2	10.7	7.8	4.3	1.7	
TITE two-stage	5.4	10.2	7.1	3.4	1.5	
IPDE-first non-TITE two-stage	5	10.2	7.9	4.7	1.9	
Mean trial duration (days)	Mean duration (days)					
TITE one-stage	1102.8					
IPDE-first non-TITE one-stage	1276.3					
TITE two-stage	1041.7					
IPDE-first non-TITE two-stage	1280.9					

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	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 24						
DLT rate	0.01	0.02	0.05	0.1	0.25	
Efficacy rate	0.05	0.08	0.2	0.5	0.5	
OBD selection (%)						
TITE one-stage	0.8	0.7	3	54.7	40.8	0
IPDE-first non-TITE one-stage	1	1.3	5.2	59.4	33.1	0
TITE two-stage	0.3	1.1	4.4	45.8	48.3	0.1
IPDE-first non-TITE two-stage	1.1	1.1	5.7	55	37	0.1
Mean administrations by dose						
TITE one-stage	3.1	3.2	4	12.4	7.4	
IPDE-first non-TITE one-stage	3.5	3.4	4.5	12.1	6.4	
TITE two-stage	3.7	3.7	5.1	10.2	7.1	
IPDE-first non-TITE two-stage	3.4	3.4	4.5	10.6	8	
Mean trial duration (days)			Mean duration (days)			
TITE one-stage			1217.4			
IPDE-first non-TITE one-stage			1321.9			
TITE two-stage			1191.5			
IPDE-first non-TITE two-stage			1332.5			

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	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 27						
DLT rate	0.11	0.17	0.3	0.5	0.62	
Efficacy rate	0.1	0.2	0.4	0.42	0.45	
OBD selection (%)						
TITE one-stage	9.1	20.1	47.1	9.8	0.6	13.3
IPDE-first non-TITE one-stage	11.2	20.8	49	12.1	0.6	6.3
TITE two-stage	8.4	24.3	40.2	10.3	1.7	15.1
IPDE-first non-TITE two-stage	10.7	24.1	49.4	9.8	0.6	5.4
Mean administrations by dose						
TITE one-stage	5.7	7.5	9.6	4.2	0.8	
IPDE-first non-TITE one-stage	7	7.8	10.3	3.6	0.6	
TITE two-stage	7.4	6.9	8.2	3.3	0.9	
IPDE-first non-TITE two-stage	6.8	7.7	9.9	3.9	1.1	
Mean trial duration (days)			Mean duration (days)			
TITE one-stage			1095.6			
IPDE-first non-TITE one-stage			1309.8			
TITE two-stage			1040.3			
IPDE-first non-TITE two-stage			1321.5			

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	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 38						
DLT rate	0.05	0.1	0.12	0.15	0.3	
Efficacy rate	0.05	0.08	0.12	0.15	0.5	
OBD selection (%)						
TITE one-stage (N/A)	N/A	N/A	N/A	N/A	N/A	N/A
IPDE-first non-TITE one-stage	4.4	5.3	9.6	12.1	64.4	4.2
TITE two-stage (N/A)	N/A	N/A	N/A	N/A	N/A	N/A
IPDE-first non-TITE two-stage	4	6.4	9.6	11.9	64	4.1
Mean administrations by dose						
TITE one-stage (N/A)	N/A	N/A	N/A	N/A	N/A	
IPDE-first non-TITE one-stage	4.7	4.3	5.2	5.5	10.2	
TITE two-stage (N/A)	N/A	N/A	N/A	N/A	N/A	
IPDE-first non-TITE two-stage	4.7	4.3	5.2	5.4	10.3	
Mean trial duration (days)			Mean duration (days)			
TITE one-stage (N/A)						-
IPDE-first non-TITE one-stage						1288.7
TITE two-stage (N/A)						-
IPDE-first non-TITE two-stage						1288.9